

RIGAKU EUV OPTICS AND DETECTOR TECHNOLOGY

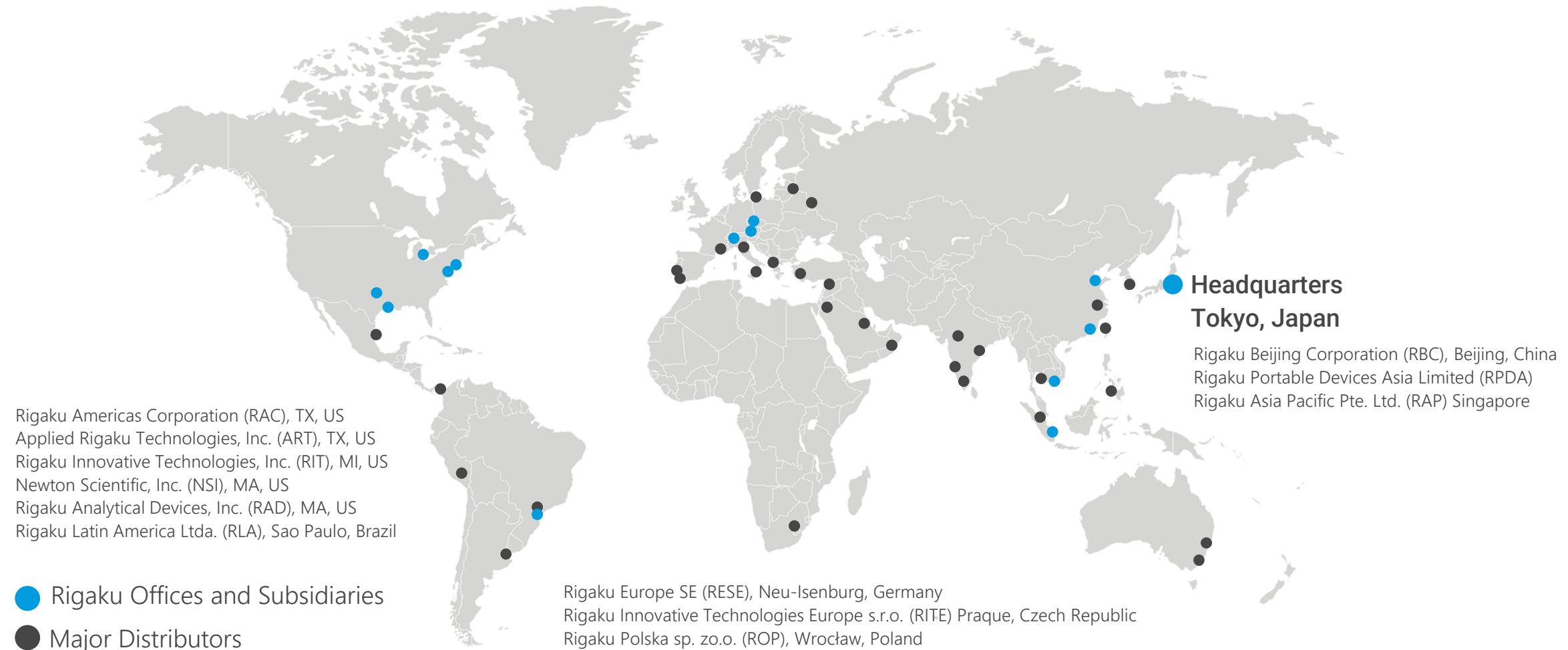
DR. PETER OBERTA



Rigaku

OUR LOCATIONS

Rigaku Offices, Subsidiaries and Major Distributors



RITE s.r.o

Rigaku established Rigaku Innovative Technologies Europe s.r.o. (RITE), 2008.
We are located in Dolní Břežany close to Prague in an academic incubator.



METROLOGY GRADED XPIN DETECTOR

ULTRA HIGH RESOLUTION EUV CAMERA

EUV COLLECTOR AND IMAGING OPTICS



xPIN:

- Metrology graded pA meter
- X-ray (air/vacuum) and EUV (pulsed/continues)
- Relative or calibrated
- Several hundreds units sold
- Various diodes available
- 150th unit sold this year.



MAX-PLANCK-GESELLSCHAFT



XSight LC camera - sCMOS:

- Lens coupled (LC) on sCMOS
- FOV: 11×11 mm @ $7 \mu\text{m}$ resolution
- FOV: 0.6×0.6 mm @ $0.5 \mu\text{m}$ resolution
- 100 fps

XSight LC camera – vacuum version

- Lens coupled (LC) on sCMOS
- FOV: 11×11 mm @ $7 \mu\text{m}$ resolution
- FOV: 0.6×0.6 mm @ $0.5 \mu\text{m}$ resolution
- 100 fps

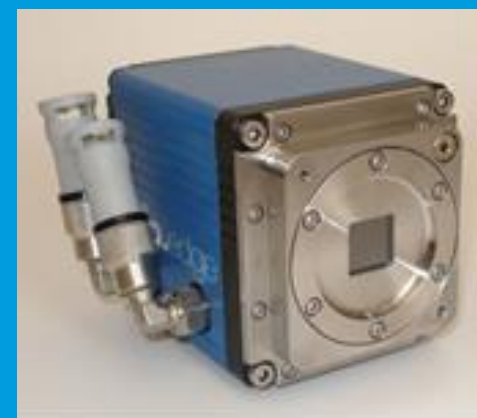


XSight FC camera:

- Fiber coupled (FC) on CCD or sCMOS
- FOV: 18×13.5 mm @ 16 μ m resolution
- FOV: 7.2×5.4 mm @ 8 μ m resolution

FC technology:

- Fiber optics coupling on alien cameras
- Magnification 1:1 or 2:1



REPLICATED AND CUSTOM MADE OPTICS

Replicated optics:

- Coating: Au, Ni, Rh, ...
- $R \geq 70\%$
- Grazing angles: $0.5^\circ - 20^\circ$
- Surface roughness: $R_a \approx 0.2 - 2 \text{ nm}$
- Dimensions: $D = 5 - 250 \text{ mm}$
 $L = 10 - 100 \text{ mm}$
- Optical shapes: ellipsoidal, parabolic, aspherical shapes upon request, Wolter
- Spectral range: EUV (60 – 200 eV)
Soft X-ray (200 – 2000 eV)
X-ray (2 - 20 keV)



CONTACT US

Rigaku ITE

+420 734 231 472

peter.oberta@rigaku.com

rigaku.com

